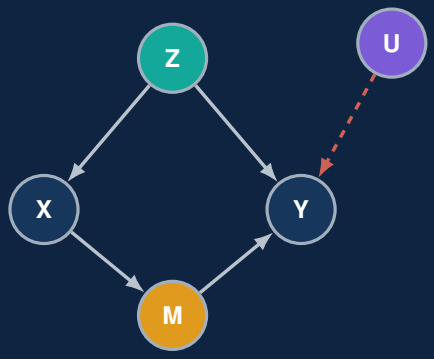


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 12

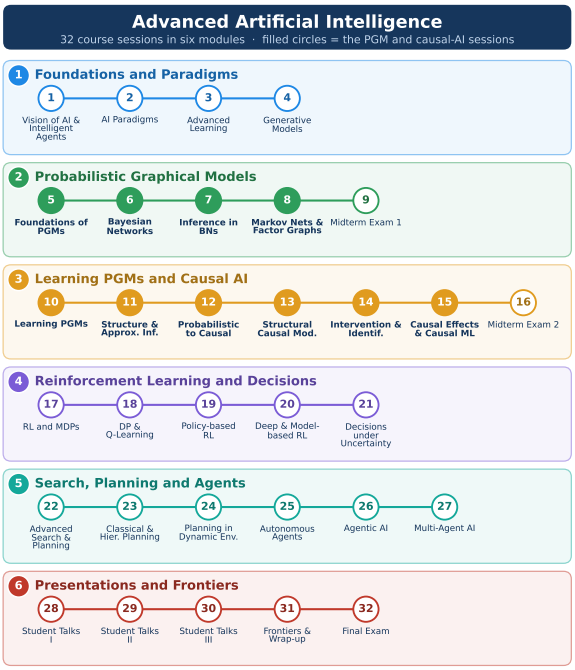
Correlation vs. Causation: The Conceptual Pivot

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 12 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s12_examples.py` for the numbers, `python/s12_figures.py` for the figures), and `questions/s12_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 12.

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SESSION 12 ADVANCED ARTIFICIAL INTELLIGENCE

Correlation vs. Causation: The Conceptual Pivot

Correlation versus causation, causal questions, causal graphs, confounding and selection bias

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Two Real Reversals
Berkeley and kidney stones: Simpson's paradox with real data.

PART II
Causal Questions & Graphs
The ladder of causation, kinds of causal question, what an arrow claims.

PART III
Confounding & Selection
Back-door paths, randomization, colliders, Berkson, survivorship, M-bias.

PART IV
Mathematical Statistics
Reversal conditions, collapsibility, randomization.

PART V
AI Connections
Agent logs, offline RL and confounded logging.

LEARNING OBJECTIVES

- Reproduce the Berkeley and kidney-stone reversals from the raw counts and explain each one with a causal graph.
- Distinguish the three rungs of the ladder of causation and classify a question as associational, interventional or counterfactual.
- State precisely what a causal DAG claims beyond a Bayesian network, and recognise confounders, mediators and colliders.
- Explain confounding and selection bias as two different graphical mechanisms, and say why "adjust for everything" fails.
- Prove when Simpson's reversal can and cannot occur and decompose a pooled difference into a within-stratum and a composition term.
- Show that randomization makes $T \perp\!\!\!\perp (Y(0), Y(1))$ and hence identifies the average treatment effect.
- Diagnose confounding and selection in logged data of LLM agents and offline RL.

Where we are. Course sessions 5–11 built the toolkit of probabilistic graphical models — conditional independence, Bayesian and Markov networks, inference, learning — all describing a *single, fixed* joint distribution P. This session (course session 12) is the hinge of the causal

block: it shows with real data that P alone cannot answer "what if we act?" questions. Structural causal models (Session 13), the do-calculus (Session 14) and causal effect estimation (Session 15) build the remedy; reinforcement learning (course sessions 17–20), agentic AI (course session 26) and multi-agent AI (course session 27) then rely on today's distinction: an agent's actions are interventions, not observations.

OPENING MOTIVATION

Every tool built in Sessions 5–11 — CI, DAGs, variable elimination, Markov networks, EM, MCMC — computes properties of a single fixed probability distribution P, generated by observing the world as it already is. Today we show, with real published data, that this entire toolkit can give you the *wrong sign* of an effect if you ask it the wrong question. Not a slightly-off estimate: the wrong sign, in data (such as the kidney-stone study below) where the comparison is used to choose a patient's treatment. This is not a footnote or a caveat to append to Sessions 5–11; it is the reason a second, distinct mathematical formalism (Sessions 13–15) is necessary at all.

PART I

Correlation vs. Causation

Three real data sets in which the pooled association has the wrong sign

PART I · REVERSALS

12.1 Two Real Reversals

Worked Example 12.1 (UC Berkeley graduate admissions, 1973)**.** In fall 1973, UC Berkeley's graduate admissions showed a troubling aggregate pattern: of 8442 male applicants about 44% were admitted, against about 35% of 4321 female applicants (Bickel, Hammel & O'Connell, *Science*, 1975). Restricting to the six largest departments (2691 men, 1835 women; full counts in Table 12.1) the gap is even wider: $1198/2691 = 44.5\%$ of men against $557/1835 = 30.4\%$ of women were admitted, a gap of 14.2 percentage points (Fig. 12.1, left). This looks like exactly the kind of evidence a sex-discrimination lawsuit would want. But when the data are broken down *by department* (middle panel and Table 12.1), something remarkable appears: in four of the six departments (A, B, D, F), women were admitted at an equal or *higher* rate than men. Department A alone admitted $89/108 = 82.4\%$ of its (relatively few) female applicants versus $512/825 = 62.1\%$ of men.

The resolution: women disproportionately applied to competitive departments with low admission rates for everyone (Department C admits $(120 + 202)/(325 + 593) = 322/918 = 35.1\%$ of all its applicants), while men disproportionately applied to less competitive departments (Department A admits $(512 + 89)/(825 + 108) = 601/933 = 64.4\%$ of all its applicants). In numbers: of the 2691 men, $825 + 560 = 1385$ applied to the two easiest departments A and B, i.e. $1385/2691 = 51.5\%$; of the 1835 women only $108 + 25 = 133$ did, i.e. $133/1835 = 7.2\%$. Each sex's pooled rate is an average of its department rates weighted by *its own* department mix, and these mixes are very different; aggregating across departments therefore mixes "where people applied" into what looks like a sex

Table 12.1. UC Berkeley 1973, the six largest departments (real data; Bickel, Hammel & O'Connell, 1975). Rates are admitted/applied. n_d = all applicants to department d, $N = \sum_d n_d = 4526$; the column n_d/N is the pooled department mix used for standardization.

Dept	Men: admitted/applied	Women: admitted/applied	n_d	n_d/N
A	$512/825 = 0.621$	$89/108 = 0.824$	933	0.206
B	$353/560 = 0.630$	$17/25 = 0.680$	585	0.129
C	$120/325 = 0.369$	$202/593 = 0.341$	918	0.203
D	$138/417 = 0.331$	$131/375 = 0.349$	792	0.175
E	$53/191 = 0.277$	$94/393 = 0.239$	584	0.129
F	$22/373 = 0.059$	$24/341 = 0.070$	714	0.158
All six	$1198/2691 = 0.445$	$557/1835 = 0.304$	4526	1

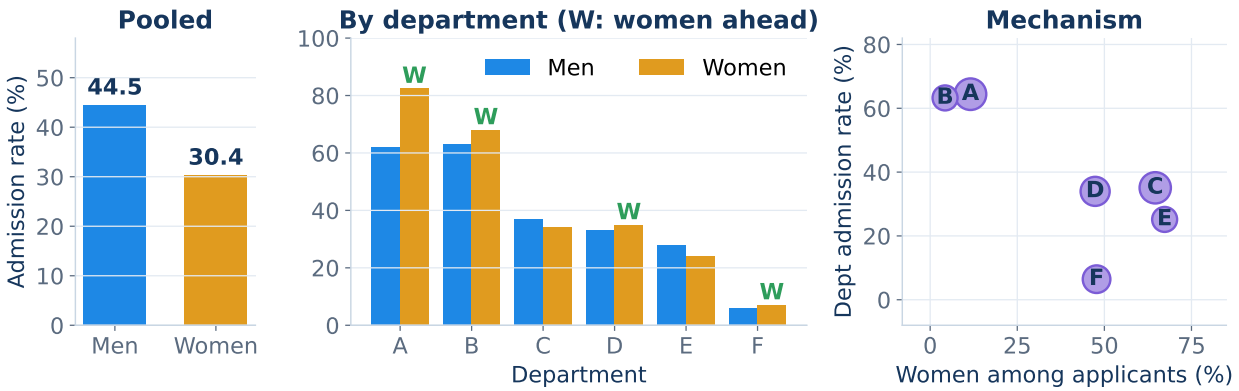


Figure 12.1. Pooled (left) vs. department-stratified (middle) admission rates, UC Berkeley 1973, six largest departments; "W" marks departments where women's rate is at least men's. Right: each department's overall admission rate against the share of women among its applicants (bubble area $\propto$ applicants). Real data (Bickel, Hammel & O'Connell, 1975).

effect at the admission stage. Note the causal direction: department choice is not a cause of sex but a *consequence* of it (sex → department → admission), a point we return to in Section 12.2.

If both sexes faced the *same* department mix — the pooled mix n_d/N of all 4526 applicants (last column of Table 12.1) — their department-specific rates would give (*direct standardization*, Section 12.9)

$$\begin{aligned}\text{men: } & 0.206(0.621) + 0.129(0.630) + 0.203(0.369) + 0.175(0.331) + 0.129(0.277) + 0.158(0.059) \\ & = 0.1279 + 0.0815 + 0.0749 + 0.0579 + 0.0358 + 0.0093 = 0.387, \\ \text{women: } & 0.206(0.824) + 0.129(0.680) + 0.203(0.341) + 0.175(0.349) + 0.129(0.239) + 0.158(0.070) \\ & = 0.1699 + 0.0879 + 0.0691 + 0.0611 + 0.0309 + 0.0111 = 0.430,\end{aligned}$$

i.e. 38.7% for men and 43.0% for women (the products are computed from unrounded rates).

PYTHON LAB · s12_examples.py · ex_berkeley()

```
aggregate:  men 1198/2691 = 0.445, women 557/1835 = 0.304
Men applying to A or B: 1385/2691 = 0.515; women:  133/1835 = 0.072
terms (n_d/N)*rate, men:  0.1279 + 0.0815 + 0.0749 + 0.0579 + 0.0358 +
0.0093
terms (n_d/N)*rate, women:  0.1699 + 0.0879 + 0.0691 + 0.0611 + 0.0309 +
0.0111
Standardised to pooled department mix:  men 0.387, women 0.430
```

Worked Example 12.2 (Kidney stone treatment, Charig et al. 1986)**.** A comparative study of two kidney-stone treatments — open surgery (Treatment A) and percutaneous nephrolithotomy (Treatment B) — reported, in aggregate, that Treatment B succeeded 82.6% of the time ($234 + 55 = 289$ successes among $270 + 80 = 350$ patients) versus 78.0% ($81 + 192 = 273$ among $87 + 263 = 350$) for Treatment A (Fig. 12.2, left), suggesting B is the better treatment. Yet stratified by stone size (Fig. 12.2, middle, using the standard tabulation by Julious & Mullee, *BMJ*, 1994): for *small* stones, A succeeds 93.1% (81/87) of the time versus B's 86.7% (234/270); for *large* stones, A succeeds 73.0% (192/263) versus B's 68.8% (55/80). Treatment A is **better in both subgroups**, yet worse overall. This is not a rounding artifact — it is a full reversal of direction.

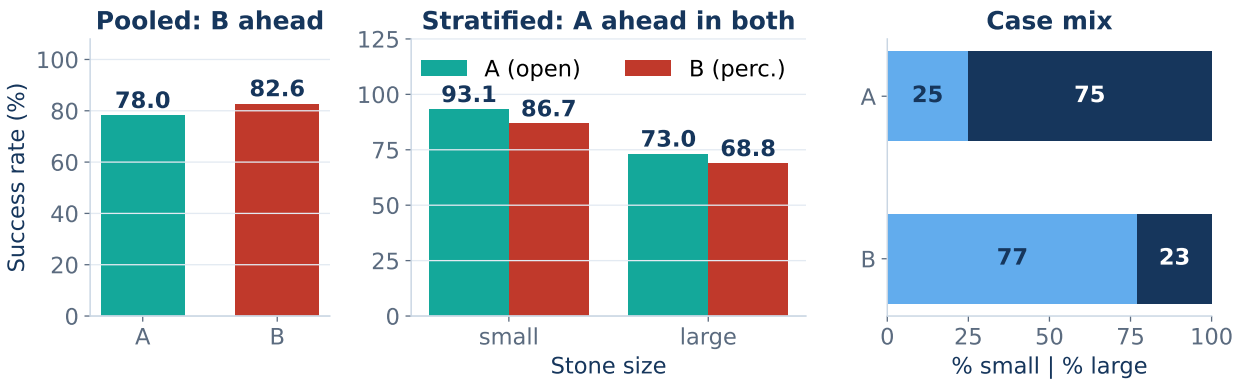


Figure 12.2. Kidney stones. Pooled (left) vs. stone-size-stratified (middle) success rates, and the case mix of each treatment (right): $263/350 = 75\%$ of A's patients had large stones, against $80/350 = 23\%$ of B's. Real data (Charig et al., 1986).

The mechanism is identical to Berkeley's: physicians preferentially used the less invasive Treatment B on *small* (easier) stones and reserved open surgery (Treatment A) for *large*, harder cases (Fig. 12.2,

right). Stone size drives both which treatment a patient received and how likely they were to recover regardless of treatment. Aggregating over stone size mixes "how severe was your case" into what looks like a treatment effect.

PYTHON LAB · s12_examples.py · ex_kidney()

```
small stones:  A 81/87 = 0.9310; B 234/270 = 0.8667
large stones:  A 192/263 = 0.7300; B 55/80 = 0.6875
aggregate A: 273/350 = 0.7800; aggregate B: 289/350 = 0.8257
```

Definition 12.1 (Simpson's paradox)**.** An association between X and Y that holds in the aggregated population reverses (or vanishes) when the population is stratified by a third variable Z.

PART I · REVERSALS

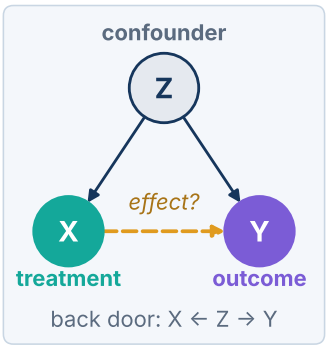
12.2 The Graphical Diagnosis

The kidney-stone study has the causal structure of Fig. 12.3: stone size Z is a common cause of the treatment X (physicians chose by stone size) and of the outcome Y (recovery). In Session 6's vocabulary, Z is the middle node of a *fork* $X \leftarrow Z \rightarrow Y$. Marginally, this fork *induces* an association between X and Y that has nothing to do with any effect of X on Y — exactly as the latent "enthusiast" class of Session 5 made two unrelated film ratings dependent. The difference now is practical: the spurious dependence sits between the treatment we must choose and the outcome we care about. The agent logs of Worked Example 12.5 have the same fork structure.

Berkeley shares the *arithmetic* of the reversal (the two groups have different distributions over Z) but not the graph: sex cannot be caused by department, so the structure is $X \rightarrow Z \rightarrow Y$ (sex influences department choice, which influences admission), possibly with a direct arrow $X \rightarrow Y$. Department is a *mediator*, not a confounder. Stratifying by department then answers a different, narrower question — is there a *direct* effect of sex on admission within the department that made the decision? — whereas the pooled comparison measures the total association. Which of the two measures "discrimination" is a causal and normative question, not a statistical one (Pearl 2009, Sec. 6.1; see the discussion question in Section 12.13).

Definition 12.2 (Confounder, informally)**.** A variable Z is a *confounder* for the relationship between X and Y if Z causally influences both X and Y. Naively comparing $P(Y \mid X = 1)$ to $P(Y \mid X = 0)$ then reflects a mixture of (i) any genuine effect of X on Y and (ii) the spurious association induced by Z.

The seemingly obvious fix — "stratify by (control for) Z" — is exactly what resolved the kidney-stone reversal, and it will be formalized properly as the *back-door criterion* in Session 14. Berkeley already shows that the fix needs a graph: stratifying by a mediator changes the question being answered. But notice we have already seen, in Session 6, that conditioning is not *always* the right move: conditioning on a **collider** ($X \rightarrow Z \leftarrow Y$, rather than a confounding fork $X \leftarrow Z \rightarrow Y$) *creates* spurious association where none existed before. A blanket rule of "control for everything you measured" is therefore not merely inelegant — it can be actively wrong, and can manufacture exactly the kind of paradox we've just resolved, in the opposite direction. This tension — when to condition, when not to — is precisely why Session 14 will insist on a checkable graphical *criterion*



Example	Z	X	Y
Kidney stones	stone size	A or B	success
Agent logs	difficulty	tool call	success

Figure 12.3. The confounding fork: Z drives both the "treatment" X and the outcome Y, opening the back-door path $X \leftarrow Z \rightarrow Y$. Right: two examples of this session with this structure (kidney stones: real data; the agent logs of Worked Example 12.5: hypothetical). Berkeley is *not* in the list: there department is a mediator (see text).

rather than a rule of thumb.

PART I · REVERSALS

12.3 Why $P(Y \mid X{=}x)$ Is the Wrong Quantity

Here is the sharpest way to state what went wrong in the kidney-stone example. The number we computed, $P(Y = 1 \mid X = x)$, answers: *"among the units we observed that happened to have $X = x$, what fraction had $Y = 1$?"* It is a fact about who ended up in each group — and who ended up in each treatment group was itself driven by stone size Z. What we actually want to know is different: *"if we were to take a patient and force X to equal x, overriding whatever process would ordinarily determine X for them, what would happen to Y?"* This is not a question about a subpopulation defined by an observed value of X; it is a question about an *action* performed on a unit, irrespective of how that unit would ordinarily have arrived at that value of X.

We introduce notation for this now, informally, and formalize it fully next session:

$$\underbrace{P(y \mid x)}_{\text{observational: who has } X{=}x?} \qquad \text{vs.} \qquad \underbrace{P\big(y \mid \mathrm{do}(x)\big)}_{\text{interventional: set } X{=}x \text{ for everyone}} . \tag{12.1}$$

For the fork of Fig. 12.3, intervening on X cuts the arrow $Z \rightarrow X$ but leaves the distribution of Z untouched, which gives (Session 13 derives this from the truncated factorization of a causal model; Session 14 generalizes it as the back-door adjustment formula)

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)\, P(z) \qquad \text{whereas} \qquad P(y \mid x) = \sum_z P(y \mid x, z)\, P(z \mid x). \tag{12.2}$$

The two differ only in the weights: $P(z)$ versus $P(z \mid x)$. Every reversal in this session lives in that difference.

> *Key Takeaway* 12.1. In general, $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$. Every session so far has computed the left-hand side. Nothing in Sessions 5–11 — no amount of more data, better inference algorithms, or more careful sampling — closes this gap, because the gap is not a statistical estimation problem. It is a conceptual one: the graphs of Sessions 5–11 (Bayesian networks

and Markov networks alike) encoded only a factorization of P, and a factorization alone does not specify what happens when a piece of the generative process is overridden rather than observed.

12.3.1 Six reasons two variables can be correlated

Confounding is only one way to obtain an association without a causal effect. Before reading $\mathrm{Cov}(X,Y) \neq 0$ as "X causes Y", six explanations must be weighed: (1) causation $X \to Y$; (2) reverse causation $Y \to X$; (3) confounding $X \leftarrow Z \to Y$; (4) selection on a common effect, $X \to S \leftarrow Y$ with only $S = 1$ observed; (5) chance (small samples, many comparisons); (6) measurement error, e.g. adjusting for a noisy proxy of Z.

Definition 12.3 (Reichenbach's common-cause principle)**.** If two variables X and Y are statistically dependent, then either X causes Y, or Y causes X, or there is a third variable Z that causes both; in the last case Z *screens off* the dependence: $X \perp\!\!\!\perp Y \mid Z$. (Reichenbach 1956; see Peters, Janzing & Schölkopf 2017, Ch. 1.)

The principle is the bridge between Sessions 5–11 and causality: dependence demands a causal explanation. Two caveats sharpen it. First, it presupposes that we look at the *whole* population; a dependence manufactured by *selection* (reason 4) has no common cause at all — the sample was filtered on a common *effect*. Second, *reverse causation* (reason 2) is indistinguishable from forward causation by $P(x, y)$ alone, as the next example shows.

Worked Example 12.3 (Reverse causation is invisible in the joint distribution)**.** *(Analytical.)* Model 1: $X \sim N(0,1)$, $Y := 0.6X + 0.8U_Y$. Model 2: $Y \sim N(0,1)$, $X := 0.6Y + 0.8U_X$, with independent standard normal noises U_X, U_Y, independent of the root variable. In Model 1, $\mathrm{Var}X = 1$, $\mathrm{Var}Y = 0.6^2\mathrm{Var}X + 0.8^2\mathrm{Var}U_Y = 0.36 + 0.64 = 1$ and $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X, 0.6X + 0.8U_Y) = 0.6\mathrm{Var}X = 0.6$; Model 2 gives the same three numbers with the roles of X and Y swapped. Both pairs are jointly Gaussian with mean 0, so both models induce the *same* bivariate normal law $N\big(0, \left(\begin{smallmatrix} 1 & 0.6 \\ 0.6 & 1 \end{smallmatrix}\right)\big)$. Yet in Model 1 setting $X = 1$ gives $Y = 0.6 \cdot 1 + 0.8U_Y$, so $\mathbb{E}[Y \mid \mathrm{do}(X = 1)] = 0.6$, while in Model 2 the equation $Y \sim N(0,1)$ does not involve X, so $\mathbb{E}[Y \mid \mathrm{do}(X = 1)] = \mathbb{E}[Y] = 0$: setting X does nothing to its own cause. No amount of observational data distinguishes them; the difference lives on rung 2 of the ladder (Section 12.4).

PYTHON LAB · `s12_examples.py` · `ex_reverse()`

```
simulation, n = 200000 draws per model, seed 7 (sample covariance matrices):
X->Y: cov = [[0.998, 0.599], [0.599, 0.997]]; Y->X: cov = [[0.999, 0.601],
[0.601, 1.002]]
```

PART II

Causal Questions and Causal Graphs

What we ask, what an arrow claims, and which variables play which role

PART II · CAUSAL QUESTIONS

12.4 Pearl's Ladder of Causation

Pearl (2009; Pearl & Mackenzie 2018) organises causal questions into three rungs (Fig. 12.4): *association* (seeing; supervised learning and LLM pre-training live here), *intervention* (doing; experiments, A/B tests and every action of an RL agent) and *counterfactuals* (imagining; attribution, regret and credit assignment). Each rung asks a strictly richer kind of question, and — this is the central message of the ladder — *information from a lower rung is in general insufficient to answer questions on a higher rung.* Worked Example 12.3 already showed two models that agree on all rung-1 queries and disagree on a rung-2 query.

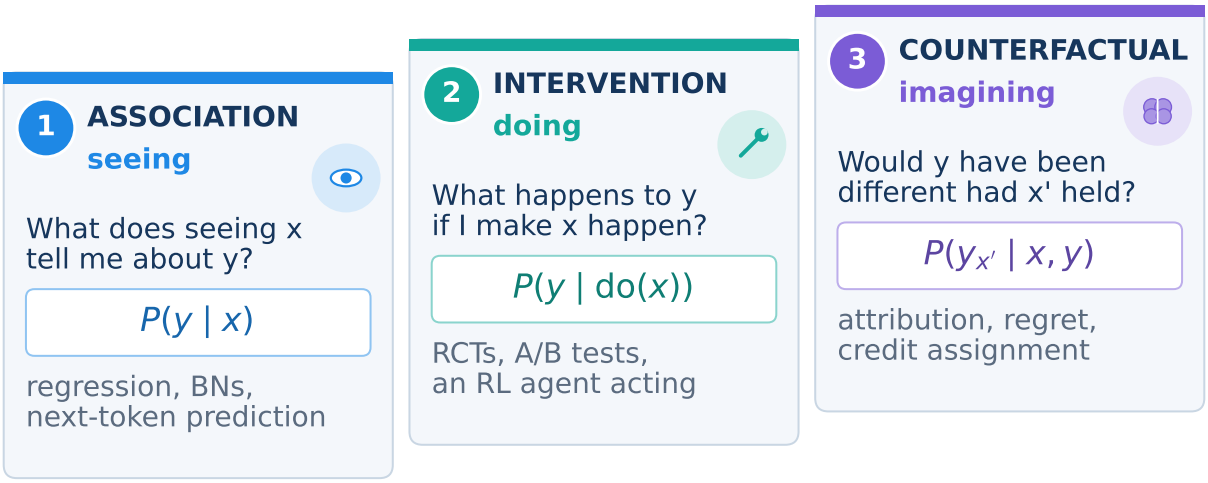


Figure 12.4. The ladder of causation. Rung 1 is the whole of Sessions 5–11; rungs 2 and 3 need a causal model (Sessions 13–15).

PART II · CAUSAL QUESTIONS

12.5 A Taxonomy of Causal Questions

"Does X cause Y?" hides several different questions: the *effect of a cause* (what would happen to Y under $\mathrm{do}(X = 1)$ versus $\mathrm{do}(X = 0)$, on average or in a subgroup), the *cause of an effect* (given that Y happened after X, would it have happened without X?), and *mediation* (how much of the effect passes through a third variable). Each will reappear with its own estimand later in the block.

PART II · CAUSAL QUESTIONS

12.6 Causal Graphs: What an Arrow Claims

In Sessions 5–6 (and in the structure learning of Session 11) a DAG was a bookkeeping device for a factorization $P(v) = \prod_i P(v_i \mid \mathrm{pa}_i)$; many DAGs (a whole Markov-equivalence class) represent the same P, and the direction of an arrow carried no meaning beyond that. A *causal* DAG makes a much stronger claim.

Definition 12.4 (Causal DAG)**.** A DAG G over variables V is a *causal DAG* if (i) an arrow $V_j \to V_i$ means that V_j is a *direct cause* of V_i relative to V: some intervention on V_j changes the distribution of V_i while all other variables in V are held fixed by intervention; and (ii) every common cause of two or more variables in V is itself in V (*causal sufficiency*), or is represented explicitly as a latent node.

Definition 12.5 (Causal Markov assumption, preview)**.** Given a causally sufficient causal DAG G, every variable is independent of its non-descendants given its direct causes: $V_i \perp \perp \mathrm{nd}(V_i) \mid \mathrm{pa}(V_i)$. Equivalently, P factorizes over G. Under an intervention $\mathrm{do}(V_k = v_k)$ the factor $P(v_k \mid \mathrm{pa}_k)$ is removed and all other factors remain *unchanged* (modularity).

This modularity is what a Bayesian network does not claim: the conditionals are *autonomous mechanisms*, so an intervention replaces one and leaves the rest intact (derived from structural equations in Session 13). As for Bayesian networks, the strongest assumptions live in the arrows that are *absent*.

12.6.1 Three roles a third variable can play

When we study the effect of X on Y, a third variable can play one of three basic structural roles (Fig. 12.5): a confounder Z (Definition 12.2), a mediator M or a collider S (defined next). The decision "should I adjust for this variable?" depends on its role, never on how strongly it correlates with X or Y.

Definition 12.6 (Mediator and collider)**.** On a path between X and Y, M is a *mediator* if the path contains $X \to M \to Y$ (M lies on a causal path from X to Y); S is a *collider* on a path if both edges of the path point into it, $\cdots \to S \leftarrow \cdots$.

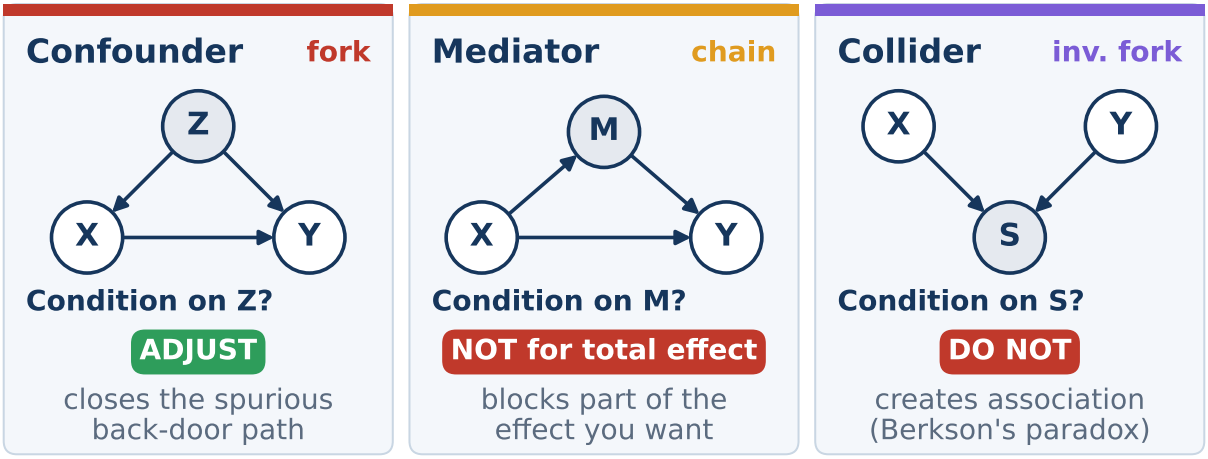


Figure 12.5. Confounder, mediator, collider. For the *total* effect of X on Y: adjust for the confounder, do not adjust for the mediator, never adjust for the collider (or any of its descendants).

PART III

Confounding and Selection Bias

Two different graphical mechanisms that make $P(y \mid x)$ lie

PART III · BIAS

12.7 Confounding

Definition 12.7 (Confounding bias; back-door path)**.** The effect of X on Y is *confounded* if $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ for some x, y. A *back-door path* from X to Y is a path that starts with an arrow *into* X, such as $X \leftarrow Z \rightarrow Y$. In a causal DAG, open back-door paths are the graphical source of confounding; Session 14 shows that blocking all of them (without opening new ones) by conditioning on a set Z yields Eq. (12.2).

The definition is causal: no property of P alone certifies the absence of confounding, and the traditional checklist ("Z associated with X and with Y, not on the causal path") is not a valid substitute (Hernán & Robins 2020, Ch. 7). Passing the checklist does not make adjustment correct: in the M-bias structure $X \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$ the pre-treatment variable M is associated with both X and Y and is not on the causal path, yet adjusting for it *creates* bias. Failing to find a variable that passes the checklist does not certify the absence of confounding either: if the common cause is unmeasured (as in Worked Example 12.6), there is no variable to check, and the data alone cannot reveal it.

12.7.1 Randomization: the gold standard

The cleanest way to remove confounding is to make X itself random: if a coin decides the treatment, nothing about the unit can influence X, so every arrow into X, and with it every back-door path, is cut. Section 12.11 states this precisely with potential outcomes: writing T for a randomized binary treatment and $Y(0), Y(1)$ for the outcomes the unit would have under each treatment, randomization gives $T \perp\!\!\!\perp (Y(0), Y(1))$ (Theorem 12.2). RCTs in medicine and A/B tests in technology are the operational form of rung 2: we *do* x instead of watching who has x.

Measurement error and proxies. Often the confounder is measured with error (an income bracket for wealth, a triage score for severity). Adjusting for a noisy proxy $W = Z + e$ removes only part of the bias — *residual confounding* — and the remainder grows as the proxy's reliability falls.

PART III · BIAS

12.8 Selection Bias

Definition 12.8 (Selection bias)**.** Let $S \in \{0, 1\}$ indicate whether a unit enters the data set. *Selection bias* occurs when the analysed distribution $P(x, y \mid S = 1)$ differs from the target $P(x, y)$ in a way that changes the association of interest; graphically, this happens when S is a descendant of Y (or of a common descendant of X and Y), so that conditioning on $S = 1$ conditions on a collider or on the outcome itself.

Berkson's paradox (collider bias). Berkson (1946) observed that two diseases that are independent in the population appear *negatively* associated among hospital patients, because a person is hospitalised if they have one *or* the other: hospitalisation is a collider. The same mechanism makes talent and effort negatively correlated among admitted students, or attractiveness and kindness among people one has dated. For example, if talent X and effort Y are independent $N(0, 1)$ and admission requires $X+Y > \sqrt{2}$, the correlation among the admitted is -0.67 (Question 5 of the question set derives it).

Survivorship bias. Selection on the outcome itself. The classic anecdote is Abraham Wald's analysis of returning WWII bombers: bullet holes were rare on the engines not because engines were never hit but because planes hit there did not return. In AI, data sets of "successful trajectories" used to evaluate or fine-tune agents, or benchmarks that silently drop crashed runs, are survivor samples.

Self-selection and M-bias. Volunteers, survey respondents and users who choose to rate an answer differ systematically from the population. A subtler trap is *M-bias*: a *pre-treatment* variable M that is a collider between two latent causes, $X \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$. The path is blocked as long as we leave M alone; adjusting for M "because it was measured before treatment" opens it.

PART IV

The Mathematics of Reversal and Bias

When reversals can happen, how large collider bias is, and why randomization works

PART IV · MATHEMATICAL STATISTICS

12.9 Simpson's Reversal as a Weighted-Average Phenomenon

Write $p_{xz} = P(Y = 1 \mid X = x, Z = z)$, $w_z^x = P(Z = z \mid X = x)$ and $p_x = P(Y = 1 \mid X = x) = \sum_z w_z^x p_{xz}$ (law of total probability). The pooled rate of each group is a weighted average of its stratum rates, *with group-specific weights*. Everything follows from that.

Theorem 12.1 (Decomposition of the pooled difference)**.** *For any discrete Z,*

$$p_1 - p_0 = \underbrace{\sum_z w_z^1 \,(p_{1z} - p_{0z})}_{\text{within-stratum effect}} + \underbrace{\sum_z (w_z^1 - w_z^0)\, p_{0z}}_{\text{composition (confounding) term}} . \tag{12.3}$$

If Z is binary, the composition term equals $(w_1^1 - w_1^0)(p_{01} - p_{00})$: the product of the X–Z association and the Z–Y association among the untreated.

Proof. $p_1 - p_0 = \sum_z w_z^1 p_{1z} - \sum_z w_z^0 p_{0z}$. Add and subtract $\sum_z w_z^1 p_{0z}$ and regroup to obtain Eq. (12.3). For binary Z, $\sum_z (w_z^1 - w_z^0) = 0$ gives $w_0^1 - w_0^0 = -(w_1^1 - w_1^0)$, so the composition term is $(w_1^1 - w_1^0)p_{01} - (w_1^1 - w_1^0)p_{00}$. □

Proposition 12.1 (No reversal without X–Z dependence)**.** *If $Z \perp\!\!\!\perp X$, then $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$. Hence if $p_{1z} \geq p_{0z}$ for every z, then $p_1 \geq p_0$, with strict inequality if some stratum with $P(z) > 0$ is strict. Consequently a Simpson reversal with binary Z requires both $w_1^1 \neq w_1^0$ and $p_{01} \neq p_{00}$, and more precisely*

$$(w_1^1 - w_1^0)(p_{01} - p_{00}) < -\sum_z w_z^1 (p_{1z} - p_{0z}) < 0.$$

Proof. If $Z \perp\!\!\!\perp X$, then $w_z^1 = w_z^0 = P(z)$, the composition term vanishes and the pooled difference is a convex combination of the stratum differences, which preserves sign. For the second statement, a reversal means every $p_{1z} - p_{0z} > 0$ (so the within term is positive) while $p_1 - p_0 < 0$; by Theorem 12.1 the composition term must be negative and exceed the within term in absolute value. □

Worked Example 12.4 (The decomposition on the kidney-stone data)**.** Apply Theorem 12.1 with $x = 1$ for Treatment A, $x = 0$ for Treatment B, $Z = 0$ for small and $Z = 1$ for large stones. The data of Example 12.2 give:

1. *Stratum rates.* $p_{10} = 81/87 = 0.9310$, $p_{00} = 234/270 = 0.8667$; $p_{11} = 192/263 = 0.7300$, $p_{01} = 55/80 = 0.6875$. Differences: $p_{10} - p_{00} = 0.0644$, $p_{11} - p_{01} = 0.0425$.
2. *Case-mix weights.* $w_0^1 = 87/350 = 0.2486$, $w_1^1 = 263/350 = 0.7514$; $w_0^0 = 270/350 = 0.7714$, $w_1^0 = 80/350 = 0.2286$.
3. *Within-stratum effect.* $0.2486(0.0644) + 0.7514(0.0425) = 0.0160 + 0.0320 = +0.0480$.
4. *Composition term* (binary form of Theorem 12.1). $(w_1^1 - w_1^0)(p_{01} - p_{00}) = (0.7514 - 0.2286)(0.6875 - 0.8667) = 0.5229 \cdot (-0.1792) = -0.0937$.
5. *Total.* $0.0480 - 0.0937 = -0.0457$, which is indeed $p_1 - p_0 = 0.7800 - 0.8257$.

Treatment A is about 4.8 points better within strata, but its case mix (three quarters large stones) costs it 9.4 points. Re-weighting both arms to the pooled mix of all 700 patients, $P(\text{small}) = (87 + 270)/700 = 357/700 = 0.51$ and $P(\text{large}) = (263 + 80)/700 = 343/700 = 0.49$ (direct standardization, i.e. the left formula of Eq. (12.2)), gives

$$\text{A: } 0.51(0.9310) + 0.49(0.7300) = 0.4748 + 0.3577 = 0.8325,$$
$$\text{B: } 0.51(0.8667) + 0.49(0.6875) = 0.4420 + 0.3369 = 0.7789,$$

a standardized advantage of $0.8325 - 0.7789 = +0.0537$ for A (computed from unrounded rates).

PYTHON LAB · s12_examples.py · ex_simpson_decomposition()

```
differences A-B = (0.0644, 0.0425); within-stratum products = (0.0160,
0.0320)
w^A_large - w^B_large = 0.5229; pB_large - pB_small = -0.1792
aggregate gap -0.0457 = within-stratum +0.0480 + composition -0.0937
standardised success:  A 0.8325, B 0.7789, difference +0.0537
```

MATHEMATICAL STATISTICS · Standardization and the sure-thing principle

Define the standardized rate $\bar{p}_x = \sum_z P(z)\, p_{xz}$. Because the weights $P(z)$ do not depend on x, Proposition 12.1 applies verbatim: if $x = 1$ is better in every stratum, it is better after standardization. This is Savage's *sure-thing principle*, and Pearl (2009, Sec. 6.1) observes that it holds for $P(y \mid \mathrm{do}(x))$ — provided Z is not affected by X — precisely because of Eq. (12.2). The paradox disappears once we ask the rung-2 question: the observational conditional p_x violates the sure-thing principle, the interventional one does not.

PART IV · MATHEMATICAL STATISTICS

12.10 Collapsibility: Confounding Is Not the Only Reason Measures Change

Definition 12.9 (Collapsibility)**.** An effect measure $\theta(p_1, p_0)$ (risk difference $p_1 - p_0$, risk ratio p_1/p_0, odds ratio $\frac{p_1/(1-p_1)}{p_0/(1-p_0)}$) is *collapsible over* Z if, whenever $Z \perp\!\!\!\perp X$ and the stratum measures are all equal to a common value θ^*, the marginal measure also equals θ^*.

Proposition 12.2 (RD is collapsible, OR is not)**.** *If $Z \perp\!\!\!\perp X$, the marginal risk difference is the $P(z)$-weighted average of the stratum risk differences, so the RD is collapsible; the RR is collapsible as well. The odds ratio is not collapsible: there are distributions with $Z \perp\!\!\!\perp X$, identical stratum odds ratios and a different marginal odds ratio.*

Proof. With $Z \perp\!\!\!\perp X$, $p_x = \sum_z P(z) p_{xz}$, so $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$, which equals θ^* if all strata do. If every stratum has $p_{1z}/p_{0z} = \theta^*$ (with $p_{0z} > 0$), then $p_1 = \sum_z P(z)\, \theta^* p_{0z} = \theta^* p_0$, so the marginal RR is θ^* too. For the odds ratio a counterexample suffices: with $Z \sim \mathrm{Bern}(1/2)$ independent of X and $P(Y{=}1 \mid X, Z)$ equal to 0.8 vs. 0.5 (treated vs. untreated) when $Z = 0$ and 0.5 vs. 0.2 when $Z = 1$, both strata have OR $= 4$ and RD $= 0.3$, while marginally $p_1 = 0.65$, $p_0 = 0.35$, RD $= 0.3$ but OR $= \frac{0.65/0.35}{0.35/0.65} = \frac{1.8571}{0.5385} = 3.45$. The underlying reason is that $p \mapsto \log \frac{p}{1-p}$ is non-linear: the logit of an average is not the average of logits. □

PART IV · MATHEMATICAL STATISTICS

12.11 Why Randomization Works

Potential outcomes (Session 15) make the argument transparent. Each unit has two potential outcomes $Y(1), Y(0)$; we observe $Y = Y(T)$ (*consistency*). The average treatment effect is $\mathrm{ATE} = \mathbb{E}[Y(1) - Y(0)]$ and the effect on the treated is $\mathrm{ATT} = \mathbb{E}[Y(1) - Y(0) \mid T = 1]$.

Theorem 12.2 (Randomization identifies the ATE). *For any assignment mechanism with* $0 < P(T = 1) < 1$,

$$\mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0] = \mathrm{ATT} + \underbrace{\big(\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0]\big)}_{\text{selection bias}}. \tag{12.4}$$

If T *is assigned by an independent randomization device, then* $T \perp\!\!\!\perp (Y(0), Y(1))$, *the selection-bias term is zero,* $\mathrm{ATT} = \mathrm{ATE}$, *and the difference in means equals the ATE. Moreover, with* n_1 *treated and* n_0 *control units drawn i.i.d. from the arms, the difference of sample means* $\hat{\tau}$ *is unbiased with* $\mathrm{Var}(\hat{\tau}) = \sigma_1^2/n_1 + \sigma_0^2/n_0$, *where* $\sigma_t^2 = \mathrm{Var}(Y(t))$.

Proof. By consistency, $\mathbb{E}[Y \mid T = t] = \mathbb{E}[Y(t) \mid T = t]$. Adding and subtracting $\mathbb{E}[Y(0) \mid T = 1]$, $\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0] = \big(\mathbb{E}[Y(1) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 1]\big) + \big(\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0]\big)$, which is Eq. (12.4). If T is produced by a device independent of everything else (e.g. a coin, $T = \mathbb{1}\{R < q\}$ with $R \perp\!\!\!\perp (Y(0), Y(1))$), then T is a measurable function of R and hence $T \perp\!\!\!\perp (Y(0), Y(1))$. Then $\mathbb{E}[Y(t) \mid T = t'] = \mathbb{E}[Y(t)]$ for all t, t', so the bias term is 0 and $\mathrm{ATT} = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \mathrm{ATE}$. For the variance: the two sample means are independent, unbiased for $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$, with variances σ_1^2/n_1 and σ_0^2/n_0 because, given $T = t$, the observed Y has the law of $Y(t)$. □

Randomization removes confounding in expectation, not selection: drop-out after randomization re-creates a collider $T \to S \leftarrow Y$.

PART V

From Observational Data to Acting Agents

Agentic AI, reinforcement learning, multi-agent systems and deep learning

PART V · AI CONNECTIONS

12.12 Why an AI Agent Must Climb the Ladder

An agent (course session 1) perceives, decides and *acts*; its action is an intervention, a rung-2 object. Whenever an AI system learns about the consequences of actions from data generated by someone else's policy, confounding, selection and non-collapsibility are all in play.

12.12.1 Agentic AI: learning from logged tool use

Worked Example 12.5 (Does the retrieval tool help? (hypothetical logs)). An LLM agent's past policy π_0 called a retrieval tool mostly on hard queries. Logs: easy queries — with tool 90/100, without 340/400; hard queries — with tool 200/400, without 40/100. Pooled success is $(90+200)/(100+400) = 290/500 = 0.580$ with the tool and $(340+40)/(400+100) = 380/500 = 0.760$ without: a naive analysis, or a fine-tuning pipeline that imitates high-reward trajectories,

would learn "don't call the tool". Within each difficulty level the tool helps: easy $90/100 = 0.90$ vs. $340/400 = 0.85$ (+5 points), hard $200/400 = 0.50$ vs. $40/100 = 0.40$ (+10 points). The log contains $100+400 = 500$ easy and 500 hard queries, so $P(\text{easy}) = 500/1000 = 0.5$; standardizing to this mix gives tool $0.5(0.90) + 0.5(0.50) = 0.700$ vs. no tool $0.5(0.85) + 0.5(0.40) = 0.625$, an estimated effect of $0.700 - 0.625 = +0.075$ — valid only if difficulty is the *only* confounder and is recorded.

PYTHON LAB · s12_examples.py · ex_agent_logs()

```
aggregate tool:  290/500 = 0.580; aggregate no_tool:  380/500 = 0.760;
P(easy) = 500/1000 = 0.50
standardised (P(easy) = 0.50):  tool 0.700, no tool 0.625, effect +0.075
```

12.12.2 Reinforcement learning: confounded offline data

In a Markov decision process (MDP) the value of an action is interventional: $Q^\pi(s,a) = \mathbb{E}[G_t \mid S_t = s, \text{do}(A_t = a)]$, where G_t is the return (the discounted sum of the rewards after time t) and all actions after A_t follow the policy π (course session 17). Online, the agent chooses a itself, so the action is randomized by the agent's own policy and $P(\cdot \mid a) = P(\cdot \mid \text{do}(a))$. Offline RL (course session 20) learns from logs produced by a behaviour policy that may have seen information the learner does not.

Worked Example 12.6 (A confounded logged bandit (hypothetical))**.** A context is good or bad with probability 1/2; the human operator who generated the logs saw it and chose arm 1 with probability 0.9 in good contexts and 0.1 in bad ones. Mean rewards: arm 1: 0.7 (good), 0.2 (bad); arm 0: 0.8 (good), 0.3 (bad) — arm 0 is better in *every* context. The learner does not observe the context.

1. *Who is in each arm.* $P(a = 1) = 0.5(0.9) + 0.5(0.1) = 0.5$; by Bayes $P(\text{good} \mid a = 1) = 0.5(0.9)/0.5 = 0.9$ and $P(\text{good} \mid a = 0) = 0.5(0.1)/0.5 = 0.1$.
2. *Naive logged means.* Arm 1: $0.9(0.7) + 0.1(0.2) = 0.63 + 0.02 = 0.65$; arm 0: $0.1(0.8) + 0.9(0.3) = 0.08 + 0.27 = 0.35$.
3. *Interventional values* (context keeps $P(\text{good}) = 0.5$): $\mathbb{E}[R \mid \text{do}(1)] = 0.5(0.7) + 0.5(0.2) = 0.45$, $\mathbb{E}[R \mid \text{do}(0)] = 0.5(0.8) + 0.5(0.3) = 0.55$.

Greedy offline policy improvement picks the worse arm, and because the confounder is unobserved no reweighting of these logs can fix it.

PYTHON LAB · s12_examples.py · ex_offpolicy()

```
P(a=1) = 0.5*0.9 + 0.5*0.1 = 0.50; P(good | a=1) = 0.90, P(good | a=0) =
0.10
naive logged means:  arm1 0.650, arm0 0.350 (arm1 = 0.63 + 0.02, arm0 = 0.08
+ 0.27)
E[R | do(a)]:  arm1 0.450, arm0 0.550 (arm1 = 0.35 + 0.10, arm0 = 0.40 +
0.15)
```

Table 12.2. Cheat sheet of key formulas.

Object	Formula
Observational vs. interventional	$P(y \mid x) = \sum_z P(y \mid x, z)P(z \mid x)$ vs. $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$
Simpson decomposition	$p_1 - p_0 = \sum_z w_z^1(p_{1z} - p_{0z}) + \sum_z (w_z^1 - w_z^0)p_{0z}$
No reversal if	$Z \perp\!\!\!\perp X$: $p_1 - p_0 = \sum_z P(z)(p_{1z} - p_{0z})$
Berkson correlation	$\mathrm{Corr}(X, Y \mid X + Y > c) = \frac{a\lambda - \lambda^2}{2 + a\lambda - \lambda^2}$, $a = c/\sqrt{2}$; $= -\frac{1}{\pi - 1}$ at $c = 0$
Omitted-variable bias	$b_{Y \cdot X} = \beta + \gamma\, \mathrm{Cov}(X, Z)/\mathrm{Var}X$
Selection-bias decomposition	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] = \mathrm{ATT} + \mathbb{E}[Y(0) \mid T{=}1] - \mathbb{E}[Y(0) \mid T{=}0]$
Randomized difference in means	unbiased for ATE, $\mathrm{Var} = \sigma_1^2/n_1 + \sigma_0^2/n_0$

PART V · AI CONNECTIONS

12.13 Summary

Three questions for discussion, then the formulas and pitfalls to keep.

THINK ABOUT IT **Berkeley's within-department analysis assumes department choice is a legitimate pathway. If departments with many women were under-funded because they attracted women, which comparison measures discrimination?**

THINK ABOUT IT **An agent's memory stores only trajectories that the user rated highly. Which arrows does the rating add to the agent's causal graph, and what does conditioning on it do?**

THINK ABOUT IT **In a randomized A/B test of two agent versions, 30% of sessions abort before the outcome is recorded, more often for version B. Is the comparison still unbiased?**

Further reading. Pearl, Glymour & Jewell (2016), *Causal Inference in Statistics: A Primer*, Ch. 1–2; Pearl (2009), *Causality*, Ch. 1 and Sec. 6.1; Hernán & Robins (2020), *Causal Inference: What If*, Ch. 1–8.

CONNECTIONS AND TAKEAWAYS

This session put the data first: with real numbers, $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ in general, and no amount of data or better inference closes that gap by itself. Two graphical mechanisms produce it — confounding (a fork) and selection (conditioning on a collider). **Session 13** supplies the missing ingredient, the structural causal model, in which the DAGs drawn since Session 6 become mechanisms that can be overridden.